PG No. 8 $D_{2\mathrm{h}}$ mmm [orthorhombic] (internal polar dipole)

* Harmonics for rank 0

$\vec{\mathbb{Q}}_0^{(1,1)}[q](A_g)$

** symmetry

1

** expression

$$\frac{\sqrt{3}Q_x x}{3}+\frac{\sqrt{3}Q_y y}{3}+\frac{\sqrt{3}Q_z z}{3}$$

* Harmonics for rank 1

$\vec{\mathbb{G}}_1^{(1,0)}[q](B_{1g})$

** symmetry

z

** expression

$$\frac{\sqrt{2}Q_x y}{2}-\frac{\sqrt{2}Q_y x}{2}$$

$\vec{\mathbb{G}}_1^{(1,0)}[q](B_{2g})$

** symmetry

y

** expression

$$-\frac{\sqrt{2}Q_x z}{2}+\frac{\sqrt{2}Q_z x}{2}$$

$\vec{\mathbb{G}}_1^{(1,0)}[q](B_{3g})$

** symmetry

x

** expression

$$\frac{\sqrt{2}Q_y z}{2}-\frac{\sqrt{2}Q_z y}{2}$$

$\vec{\mathbb{Q}}_1^{(1,-1)}[q](B_{1u})$

** symmetry

z

** expression

Q_z

$\vec{\mathbb{Q}}_1^{(1,1)}[q](B_{1u})$

** symmetry

z

** expression

$$\frac{3\sqrt{10}Q_x xz}{10}+\frac{3\sqrt{10}Q_y yz}{10}-\frac{\sqrt{10}Q_z\left(x^2+y^2-2z^2\right)}{10}$$

$\vec{\mathbb{Q}}_1^{(1,-1)}[q](B_{2u})$

** symmetry

y

** expression

Q_y

$\vec{\mathbb{Q}}_1^{(1,1)}[q](B_{2u})$

** symmetry

y

** expression

$$\frac{3\sqrt{10}Q_x xy}{10}-\frac{\sqrt{10}Q_y\left(x^2-2y^2+z^2\right)}{10}+\frac{3\sqrt{10}Q_z yz}{10}$$

$\vec{\mathbb{Q}}_1^{(1,-1)}[q](B_{3u})$

** symmetry

x

** expression

Q_x

$\vec{\mathbb{Q}}_1^{(1,1)}[q](B_{3u})$
** symmetry

x

** expression

$$\frac{\sqrt{10}Q_x\left(2x^2-y^2-z^2\right)}{10}+\frac{3\sqrt{10}Q_yxy}{10}+\frac{3\sqrt{10}Q_zxz}{10}$$

* Harmonics for rank 2
$\vec{\mathbb{Q}}_2^{(1,-1)}[q](A_g,1)$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{6}Q_xx}{6}-\frac{\sqrt{6}Q_yy}{6}+\frac{\sqrt{6}Q_zz}{3}$$

$\vec{\mathbb{Q}}_2^{(1,-1)}[q](A_g,2)$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{2}Q_xx}{2}-\frac{\sqrt{2}Q_yy}{2}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](A_g,1)$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{21}Q_xx\left(x^2+y^2-4z^2\right)}{14}-\frac{\sqrt{21}Q_yy\left(x^2+y^2-4z^2\right)}{14}-\frac{\sqrt{21}Q_zz\left(3x^2+3y^2-2z^2\right)}{14}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](A_g,2)$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{7}Q_xx\left(3x^2-7y^2-2z^2\right)}{14}+\frac{\sqrt{7}Q_yy\left(7x^2-3y^2+2z^2\right)}{14}+\frac{5\sqrt{7}Q_zz\left(x-y\right)\left(x+y\right)}{14}$$

$\vec{\mathbb{Q}}_2^{(1,-1)}[q](B_{1g})$
** symmetry

$\sqrt{3}xy$

** expression

$$\frac{\sqrt{2}Q_xy}{2}+\frac{\sqrt{2}Q_yx}{2}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](B_{1g})$
** symmetry

$\sqrt{3}xy$

** expression

$$\frac{\sqrt{7}Q_xy\left(4x^2-y^2-z^2\right)}{7}-\frac{\sqrt{7}Q_yx\left(x^2-4y^2+z^2\right)}{7}+\frac{5\sqrt{7}Q_zxyz}{7}$$

$\vec{\mathbb{Q}}_2^{(1,-1)}[q](B_{2g})$
** symmetry

$$\sqrt{3}xz$$

** expression

$$\frac{\sqrt{2}Q_x z}{2} + \frac{\sqrt{2}Q_z x}{2}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](B_{2g})$
** symmetry

$$\sqrt{3}xz$$

** expression

$$\frac{\sqrt{7}Q_x z\left(4x^2 - y^2 - z^2\right)}{7} + \frac{5\sqrt{7}Q_y xyz}{7} - \frac{\sqrt{7}Q_z x\left(x^2 + y^2 - 4z^2\right)}{7}$$

$\vec{\mathbb{Q}}_2^{(1,-1)}[q](B_{3g})$
** symmetry

$$\sqrt{3}yz$$

** expression

$$\frac{\sqrt{2}Q_y z}{2} + \frac{\sqrt{2}Q_z y}{2}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](B_{3g})$
** symmetry

$$\sqrt{3}yz$$

** expression

$$\frac{5\sqrt{7}Q_x xyz}{7} - \frac{\sqrt{7}Q_y z\left(x^2 - 4y^2 + z^2\right)}{7} - \frac{\sqrt{7}Q_z y\left(x^2 + y^2 - 4z^2\right)}{7}$$

$\vec{\mathbb{G}}_2^{(1,0)}[q](A_u, 1)$
** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$\frac{\sqrt{6}Q_x yz}{2} - \frac{\sqrt{6}Q_y xz}{2}$$

$\vec{\mathbb{G}}_2^{(1,0)}[q](A_u, 2)$
** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{2}Q_x yz}{2} + \frac{\sqrt{2}Q_y xz}{2} - \sqrt{2}Q_z xy$$

$\vec{\mathbb{G}}_2^{(1,0)}[q](B_{1u})$
** symmetry

$$\sqrt{3}xy$$

** expression

$$-\frac{\sqrt{2}Q_x xz}{2} + \frac{\sqrt{2}Q_y yz}{2} + \frac{\sqrt{2}Q_z\left(x - y\right)\left(x + y\right)}{2}$$

$\vec{\mathbb{G}}_2^{(1,0)}[q](B_{2u})$
** symmetry

$$\sqrt{3}xz$$

** expression

$$\frac{\sqrt{2}Q_x xy}{2} - \frac{\sqrt{2}Q_y\left(x - z\right)\left(x + z\right)}{2} - \frac{\sqrt{2}Q_z yz}{2}$$

$\vec{\mathbb{G}}_2^{(1,0)}[q](B_{3u})$
** symmetry

$$\sqrt{3}yz$$

** expression

$$\frac{\sqrt{2}Q_x\left(y-z\right)\left(y+z\right)}{2}-\frac{\sqrt{2}Q_yxy}{2}+\frac{\sqrt{2}Q_zxz}{2}$$

* Harmonics for rank 3
$\vec{\mathbb{G}}_3^{(1,0)}[q](A_g)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{5}Q_xx\left(y-z\right)\left(y+z\right)}{2}-\frac{\sqrt{5}Q_yy\left(x-z\right)\left(x+z\right)}{2}+\frac{\sqrt{5}Q_zz\left(x-y\right)\left(x+y\right)}{2}$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](B_{1g},1)$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{3}Q_xy\left(x^2+y^2-4z^2\right)}{4}+\frac{\sqrt{3}Q_yx\left(x^2+y^2-4z^2\right)}{4}$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](B_{1g},2)$
** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{5}Q_xy\left(x^2-y^2+2z^2\right)}{4}-\frac{\sqrt{5}Q_yx\left(x^2-y^2-2z^2\right)}{4}-\sqrt{5}Q_zxyz$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](B_{2g},1)$
** symmetry

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

** expression

$$\frac{\sqrt{3}Q_xz\left(x^2-4y^2+z^2\right)}{4}-\frac{\sqrt{3}Q_zx\left(x^2-4y^2+z^2\right)}{4}$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](B_{2g},2)$
** symmetry

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

** expression

$$\frac{\sqrt{5}Q_xz\left(x^2+2y^2-z^2\right)}{4}-\sqrt{5}Q_yxyz-\frac{\sqrt{5}Q_zx\left(x^2-2y^2-z^2\right)}{4}$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](B_{3g},1)$
** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

** expression

$$\frac{\sqrt{3}Q_yz\left(4x^2-y^2-z^2\right)}{4}-\frac{\sqrt{3}Q_zy\left(4x^2-y^2-z^2\right)}{4}$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](B_{3g},2)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

** expression

$$-\sqrt{5}Q_x xyz + \frac{\sqrt{5}Q_y z\left(2x^2+y^2-z^2\right)}{4} + \frac{\sqrt{5}Q_z y\left(2x^2-y^2+z^2\right)}{4}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](A_u)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$Q_x yz + Q_y xz + Q_z xy$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](A_u)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{15}Q_x yz\left(6x^2-y^2-z^2\right)}{6} - \frac{\sqrt{15}Q_y xz\left(x^2-6y^2+z^2\right)}{6} - \frac{\sqrt{15}Q_z xy\left(x^2+y^2-6z^2\right)}{6}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](B_{1u},1)$
** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$-\frac{\sqrt{15}Q_x xz}{5} - \frac{\sqrt{15}Q_y yz}{5} - \frac{\sqrt{15}Q_z\left(x^2+y^2-2z^2\right)}{10}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](B_{1u},2)$
** symmetry

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$Q_x xz - Q_y yz + \frac{Q_z\left(x-y\right)\left(x+y\right)}{2}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](B_{1u},1)$
** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$-\frac{5Q_x xz\left(3x^2+3y^2-4z^2\right)}{12} - \frac{5Q_y yz\left(3x^2+3y^2-4z^2\right)}{12} + \frac{Q_z\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](B_{1u},2)$
** symmetry

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{15}Q_x xz\left(5x^2-9y^2-2z^2\right)}{12} + \frac{\sqrt{15}Q_y yz\left(9x^2-5y^2+2z^2\right)}{12} - \frac{\sqrt{15}Q_z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](B_{2u},1)$
** symmetry

$$-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$$

** expression

$$-\frac{\sqrt{15}Q_x xy}{5} - \frac{\sqrt{15}Q_y\left(x^2-2y^2+z^2\right)}{10} - \frac{\sqrt{15}Q_z yz}{5}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](B_{2u},2)$

** symmetry

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

** expression

$$-Q_xxy-\frac{Q_y\left(x-z\right)\left(x+z\right)}{2}+Q_zyz$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](B_{2u},1)$
** symmetry

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

** expression

$$-\frac{5Q_xxy\left(3x^2-4y^2+3z^2\right)}{12}+\frac{Q_y\left(3x^4-24x^2y^2+6x^2z^2+8y^4-24y^2z^2+3z^4\right)}{12}-\frac{5Q_zyz\left(3x^2-4y^2+3z^2\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](B_{2u},2)$
** symmetry

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

** expression

$$-\frac{\sqrt{15}Q_xxy\left(5x^2-2y^2-9z^2\right)}{12}+\frac{\sqrt{15}Q_y\left(x-z\right)\left(x+z\right)\left(x^2-6y^2+z^2\right)}{12}-\frac{\sqrt{15}Q_zyz\left(9x^2+2y^2-5z^2\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](B_{3u},1)$
** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

** expression

$$\frac{\sqrt{15}Q_x\left(2x^2-y^2-z^2\right)}{10}-\frac{\sqrt{15}Q_yxy}{5}-\frac{\sqrt{15}Q_zxz}{5}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](B_{3u},2)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

** expression

$$\frac{Q_x\left(y-z\right)\left(y+z\right)}{2}+Q_yxy-Q_zxz$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](B_{3u},1)$
** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

** expression

$$\frac{Q_x\left(8x^4-24x^2y^2-24x^2z^2+3y^4+6y^2z^2+3z^4\right)}{12}+\frac{5Q_yxy\left(4x^2-3y^2-3z^2\right)}{12}+\frac{5Q_zxz\left(4x^2-3y^2-3z^2\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](B_{3u},2)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

** expression

$$\frac{\sqrt{15}Q_x\left(y-z\right)\left(y+z\right)\left(6x^2-y^2-z^2\right)}{12}-\frac{\sqrt{15}Q_yxy\left(2x^2-5y^2+9z^2\right)}{12}+\frac{\sqrt{15}Q_zxz\left(2x^2+9y^2-5z^2\right)}{12}$$